

BUILDING WEBAGENTS

FOR REAL-WORLD TASKS

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Mellon
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Problem Statement



- Build a generalist web agent for any given website, can follow language instructions and carry out corresponding task
- Multimodal Mind2Web dataset to train
 - 2000 open ended tasks, 127 websites, 31 domains



- To demonstrate decent level of performance of LLM's for below tasks such as Flight booking, ordering a book, social media engagement
- Integrating with Playwright to launch website and let the LLM to complete the task

Mind2Web: Two-Stage Architecture & Metrics

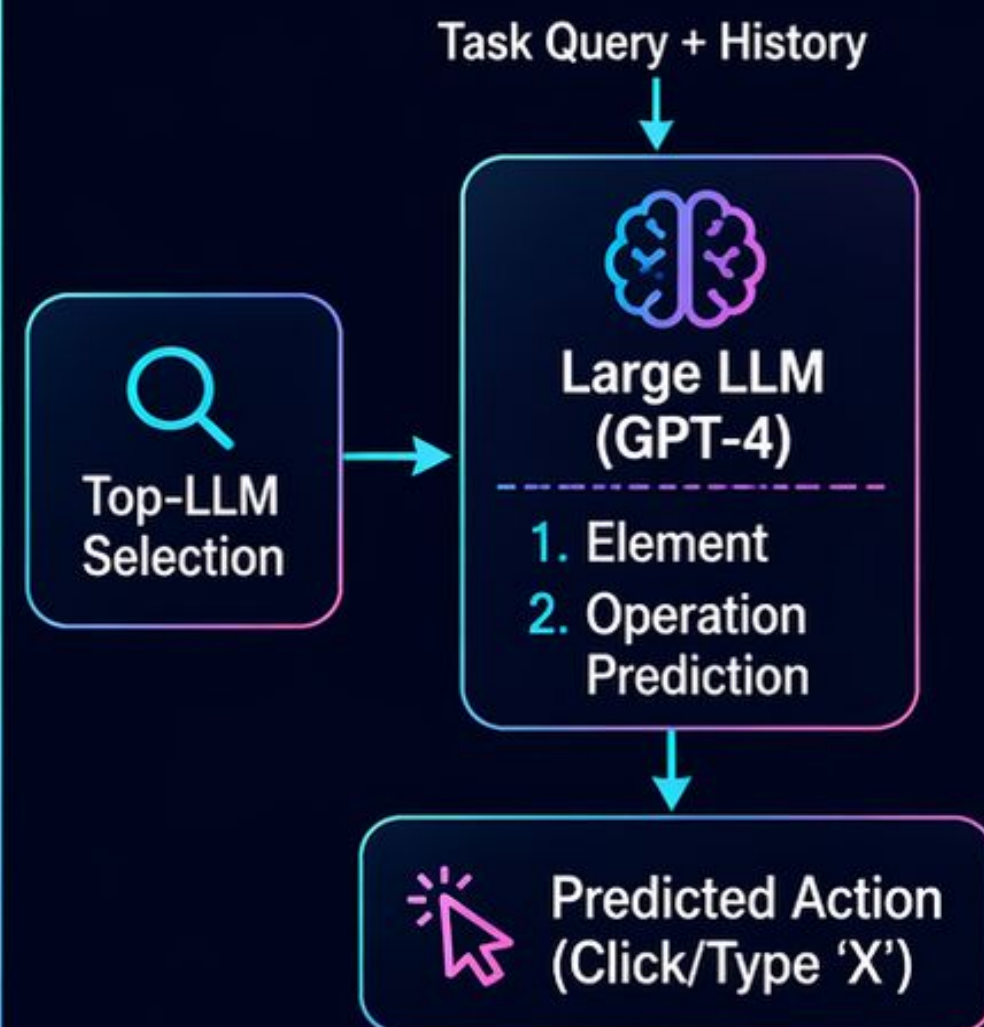
Stage 1: Candidate Generation (Filter)



Metrics:

- Recall@20: 95%

Stage 2: Action Prediction (LLM Brain)



Metrics

- Element Accuracy: 55.1%
- Operation F1: 75.7%
- Step Success Rate: 52.0%
- Task Success Rate: 5.2%

Key Insight



1 Narrow down to
top@K DOM elements



2 Cross-Encoder
(DeBERTa)



By Using II Stage Pipeline, I am able to first narrow down to top@K DOM elements with Cross-Encoder such as DeBERTa. This makes interacting with complex websites manageable so that LLM such as GPT can accurately predict next action for given task

Promising Trends/Results

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Recall@20

96%

Mean Reciprocal Rank

7.99

in Stage I



F1 score

0.46

Element Accuracy

0.22

for Stage II

Limitations



- Playwright Integration to work on travel domain site like www.kayak.com instead of generic www.google.com



- DeBERTa training instability causing 200 Nan which eventually leading to Invalid predicted action by Flan T5 & GPT-4



- Library version compatibility, Depending Issues - Pin down the library version



- Memory Issue(skip some vision,raw HTML elements) - Quantization, host on vLLM

Next Steps



- Playwright, Deployment in Production



- www.kayak.com vs www.google.com



- Improve Re-ranking Quality



- Performance/Success Rate of Web Agent is not substantial, Needs to be improved



- Stability of Web Agent like avoiding NaN Issues



- Handling more complex tasks

Contd...



- DPO(Direct Preference Optimization) - +ve example, -ve example



- Replace DeBERTa with RoBERTa



- Quantization
 - QLORA, Fine Tuning with few shots, Zero shots,



- RAG (Retrieval Augment Generation),



- storing GGUF - GPT Generated Unified Format,



- AWQ (Activation Aware Weight Quantization)

Overall Clarity



- Web Agent
- II stage Pipeline, Stage I filters, Stage II can interact with complex websites
- Recall@20 96%,
recall(95% at top 20)

References

- ▶ Mind2Web Paper - <https://arxiv.org/abs/2306.06070>
- 👤 Project website <https://osu-nlp-group.github.io/Mind2Web/>
- 💬 <https://huggingface.co/datasets/osunlp/Multimodal-Mind2Web>

Q&A

